

Bradley-Terry Problems from Agresti (2013)

Angus Scroggie

20-08-2025

These problems are from *Categorical Data Analysis* (3rd ed.) by Alan Agresti (2013). We focus on those about the Bradley-Terry (BT) model for paired comparisons, which can be found in Ch. 11.6. This is mainly for my own understanding of the model for my Honours project at UQ (2025-2026). Note that there is a companion website to the text which has a file with short solutions to selected problems (in particular, 11.16, 17, 43, and 45).

We will use the `BradleyTerry2` package by Turner & Firth (2025) to fit the model when necessary.

```
library("BradleyTerry2")
```

Applications

11.16

First, let's input the given data matrix X , where X_{ij} denotes the number of times competitor i beat j .

```
drink_names <- c("Coke", "Pepsi", "Classic Coke")
drink_matrix <- matrix(c(
  0, 29, 31,
  20, 0, 28,
  19, 19, 0
), nrow = 3, byrow = TRUE)
colnames(drink_matrix) <- drink_names
rownames(drink_matrix) <- drink_names
drink_matrix
```

```
##           Coke Pepsi Classic Coke
## Coke           0   29           31
## Pepsi          20    0           28
## Classic Coke   19   19           0
```

Convert to binomial frequencies.

```
drink_freq <- countsToBinomial(drink_matrix)
names(drink_freq)[1:2] = c("drink1", "drink2")
drink_freq
```

```
##   drink1      drink2 win1 win2
## 1   Coke        Pepsi   29   20
## 2   Coke Classic Coke   31   19
## 3   Pepsi Classic Coke   28   19
```

Fit the simple BT model. Note that the parameter estimate for *Coke* is set to zero for identifiability, but we can re-specify so that the *Classic Coke* parameter is zero instead (in line with Agresti's solution).

```

drink_model <- BTm(cbind(win1, win2), drink1, drink2, data = drink_freq)
drink_model_up <- update(drink_model, refcat = "Classic Coke")
summary(drink_model_up)

##
## Call:
## BTm(outcome = cbind(win1, win2), player1 = drink1, player2 = drink2,
##      refcat = "Classic Coke", data = drink_freq)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## ..Coke       0.5796     0.2399   2.416  0.0157 *
## ..Pepsi      0.2959     0.2399   1.233  0.2175
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6.30488  on 3  degrees of freedom
## Residual deviance: 0.28271  on 1  degrees of freedom
## AIC: 17.193
##
## Number of Fisher Scoring iterations: 3
drink_pvalue <- 1 - pchisq(drink_model_up$deviance, drink_model_up$df.residual)

```

The fit seems good, with small deviance ($D = 0.283$) and evidence that the fit is adequate ($p = 0.595$). It seems there is evidence of a preference for *Coke* over *Classic Coke* ($p \approx 0.016$). A ranking can be easily produced.

```

drink_ranking <- BTabilities(drink_model_up)
drink_ranking

```

```

##           ability      s.e.
## Coke       0.5795596 0.2398730
## Pepsi      0.2959017 0.2399236
## Classic Coke 0.0000000 0.0000000

```

We also want to compare *Coke* and *Pepsi*. The estimated probability that *Coke* is preferred is

```
exp(drink_ranking[1]) / (exp(drink_ranking[1]) + exp(drink_ranking[2]))
```

```
## [1] 0.5704428
```

and the sample proportion is

```
drink_matrix[1,2] / (drink_matrix[1,2] + drink_matrix[2,1])
```

```
## [1] 0.5918367
```

which are very similar results. It seems *Coke* is preferred to *Pepsi* a little over half the time, on average.

11.17

This problem is actually done in the vignette for `BradleyTerry2` and the data is included in the package itself. The following is largely based on the vignette example.

```

data("citations", package = "BradleyTerry2")
t(citations)

```

```
##           cited
## citing      Biometrika Comm Statist JASA JRSS-B
## Biometrika      714          33 320 284
## Comm Statist      730          425 813 276
## JASA              498          68 1072 325
## JRSS-B           221          17 142 188
```

Convert to binomial frequencies for ‘wins’ and ‘losses’ of each journal. For the comparison of pair (A, B) , we are supposing that A wins if A is cited by B , and vice versa. In the simple model, we ignore self-citations since the model parameters are relative and the interpretation doesn’t really make sense in our framework (A defeats itself).

```
citations.sf = countsToBinomial(citations)
names(citations.sf)[1:2] = c("journal1", "journal2")
citations.sf
```

```
##      journal1      journal2 win1 win2
## 1 Biometrika Comm Statist 730 33
## 2 Biometrika      JASA 498 320
## 3 Biometrika      JRSS-B 221 284
## 4 Comm Statist      JASA 68 813
## 5 Comm Statist      JRSS-B 17 276
## 6      JASA      JRSS-B 142 325
```

Now fitting the BT model. Note that one parameter (corresponding to *Biometrika*) is set to zero for identifiability; again, let’s set this for *JRSS-B* to align with Agresti’s results.

```
cite_model <- BTm(cbind(win1, win2), journal1, journal2, data = citations.sf)
cite_model_up <- update(cite_model, refcat = "JRSS-B")
summary(cite_model_up)
```

```
##
## Call:
## BTm(outcome = cbind(win1, win2), player1 = journal1, player2 = journal2,
##      refcat = "JRSS-B", data = citations.sf)
##
## Coefficients:
##      Estimate Std. Error z value Pr(>|z|)
## ..Biometrika -0.26895    0.07083 -3.797 0.000146 ***
## ..Comm Statist -3.21803    0.11230 -28.655 < 2e-16 ***
## ..JASA        -0.74852    0.07294 -10.262 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1925.2329 on 6 degrees of freedom
## Residual deviance: 4.2934 on 3 degrees of freedom
## AIC: 46.394
##
## Number of Fisher Scoring iterations: 4
```

```
cite_pvalue <- 1 - pchisq(cite_model_up$deviance, cite_model_up$df.residual)
```

There’s evidence that the fit is adequate ($p = 0.231$). The prestige ranking is, in descending order,

```
cite_ranking <- BTabilities(cite_model_up)
cite_ranking[order(cite_ranking[, 1], decreasing = TRUE), ]
```

```
##              ability      s.e.
## JRSS-B         0.0000000 0.0000000
## Biometrika    -0.2689541 0.07082997
## JASA          -0.7485238 0.07294356
## Comm Statist -3.2180266 0.11230134
```

The estimated probability that a *Comm Statist* article cites a *JRSS-B* article, for a random citation involving the two journals, is given by

```
exp(cite_ranking[4]) / (exp(cite_ranking[2]) + exp(cite_ranking[4]))
```

```
## [1] 0.961507
```

11.18

Insert the data and convert to binomial frequencies.

```
tennis_names <- c("Nadal", "Federer", "Murray", "Roddick", "Djokovic")
tennis_matrix <- matrix(c(
  0, 5, 6, 2, 6,
  2, 0, 4, 5, 7,
  3, 4, 0, 2, 1,
  1, 0, 1, 0, 4,
  7, 7, 2, 1, 0
), nrow = 5, byrow = TRUE)
colnames(tennis_matrix) <- tennis_names
rownames(tennis_matrix) <- tennis_names
tennis_matrix
```

```
##      Nadal Federer Murray Roddick Djokovic
## Nadal      0      5      6      2      6
## Federer    2      0      4      5      7
## Murray     3      4      0      2      1
## Roddick    1      0      1      0      4
## Djokovic   7      7      2      1      0
```

```
tennis_freq <- countsToBinomial(tennis_matrix)
names(tennis_freq)[1:2] = c("player1", "player2")
tennis_freq
```

```
##   player1 player2 win1 win2
## 1  Nadal  Federer    5    2
## 2  Nadal  Murray    6    3
## 3  Nadal  Roddick    2    1
## 4  Nadal  Djokovic    6    7
## 5  Federer Murray    4    4
## 6  Federer Roddick    5    0
## 7  Federer Djokovic    7    7
## 8  Murray  Roddick    2    1
## 9  Murray  Djokovic    1    2
## 10 Roddick Djokovic    4    1
```

Fit the simple BT model. We leave the default choice of zero parameter, which is for Nadal.

```
tennis_model <- BTm(cbind(win1, win2), player1, player2, data = tennis_freq)
tennis_model_up <- update(tennis_model)
```

```
summary(tennis_model_up)
```

```
##
## Call:
## BTm(outcome = cbind(win1, win2), player1 = player1, player2 = player2,
##      data = tennis_freq)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## ..Federer    -0.2322     0.4543  -0.511   0.609
## ..Murray     -0.4855     0.4900  -0.991   0.322
## ..Roddick    -0.7716     0.5998  -1.286   0.198
## ..Djokovic   -0.3015     0.4267  -0.707   0.480
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 12.303  on 10  degrees of freedom
## Residual deviance: 10.137  on  6  degrees of freedom
## AIC: 38.439
##
## Number of Fisher Scoring iterations: 4
tennis_pvalue <- 1 - pchisq(tennis_model_up$deviance, tennis_model_up$df.residual)
```

There is evidence of an adequate fit ($p = 0.119$). The player ranking is hence

```
tennis_ranking <- BTabilities(tennis_model_up)
tennis_ranking[order(tennis_ranking[, 1], decreasing = TRUE), ]
```

```
##              ability      s.e.
## Nadal         0.0000000 0.0000000
## Federer      -0.2322090 0.4542769
## Djokovic     -0.3014634 0.4266927
## Murray       -0.4855277 0.4899562
## Roddick      -0.7715631 0.5997769
```

As suggested in the hint, the model is probably not realistic. There are a variety of factors not considered in the simple Bradley-Terry model, such as weather conditions, court surfaces, and home advantage effects. The time frame is probably short enough for time-varying effects to not matter too much. A more generalised model could account for these factors (covariates, order effects).

The estimated probability of Nadal beating Federer is

```
exp(tennis_ranking[1]) / (exp(tennis_ranking[1]) + exp(tennis_ranking[2]))
```

```
## [1] 0.5577928
```

and the sample proportion is

```
tennis_matrix[1,2] / (tennis_matrix[1,2] + tennis_matrix[2,1])
```

```
## [1] 0.7142857
```

which is quite different to the model estimate. Standard errors for the predicted probabilities are provided by the `predict.BTm` method. Note: I can't seem to find any good documentation on the output of this function, but presumably the order of the probabilities in the output matrix matches that of the input binomial frequencies. A 90% confidence interval for the probability is hence

```

preds <- predict(tennis_model_up, type = "response", se.fit = TRUE)
prob_nf <- preds$fit[1]
margin_nf <- qnorm(0.95) * preds$se.fit[1]
c(prob_nf - margin_nf, prob_nf + margin_nf)

```

```

##           1           1
## 0.3734838 0.7421018

```

or 0.558 ± 0.184 which includes the sample proportion.

Theory and Methods

11.43

We have

$$\begin{aligned}
\log(\pi_{ac}/\pi_{ca}) &= \beta_a - \beta_c \\
&= (\beta_a - \beta_b) + (\beta_b - \beta_c) \\
&= \log(\pi_{ab}/\pi_{ba}) + \log(\pi_{bc}/\pi_{cb})
\end{aligned}$$

as desired. From this result, it is not possible for the probability estimates to give a cyclic preference of a, b, c . Consider by counterexample

$$\begin{aligned}
\pi_{ab} > \pi_{ba} &\Rightarrow \beta_a > \beta_b \\
\pi_{bc} > \pi_{cb} &\Rightarrow \beta_b > \beta_c \\
\pi_{ca} > \pi_{ac} &\Rightarrow \beta_c > \beta_a
\end{aligned}$$

which yields a contradiction by the transitivity of the inequality.

To show the need for identifiability, we can translate each $\beta_i^* = \beta_i - c$ where $c \in \mathbb{R}$ for each $i = 1, \dots, I$. Then the model is specified by

$$\begin{aligned}
\pi_{ab}^* &= \frac{\exp \beta_a^*}{\exp \beta_a^* + \exp \beta_b^*} \\
&= \frac{\exp(\beta_a - c)}{\exp(\beta_a - c) + \exp(\beta_b - c)} \\
&= \frac{\exp(-c) \exp \beta_a}{\exp(-c) \exp \beta_a + \exp(-c) \exp \beta_b} \\
&= \frac{\exp \beta_a}{\exp \beta_a + \exp \beta_b} \\
&= \pi_{ab}
\end{aligned}$$

11.44

The probability that team i beats team j , where i is home and j is away, is now

$$\pi_{ij}^* = \frac{\exp \beta_{Hi}}{\exp \beta_{Hi} + \exp \beta_{Aj}}$$

and the log-odds version of the model becomes

$$\log \frac{\pi_{ij}^*}{\pi_{ji}^*} = \beta_{Hi} - \beta_{Aj}$$

Set $\beta_{Hi} = \beta_{Ai} + c$ for each $i = 1, \dots, I$. When $c = 0$, the home and away parameters are equal for each competitor, which is just the original Bradley-Terry model. When $c > 0$, then we have the home advantage model described in the text.

$$\log \frac{\pi_{ij}^*}{\pi_{ji}^*} = (\beta_{Hi} + c) - (\beta_{Hj} - c) = 2c + \beta_{Hi} - \beta_{Hj}$$

[TODO fit the model and compare]

11.45

We want the likelihood for the simple BT model. Recall that the number of times i beats j , denoted n_{ij} , is distributed with

$$n_{ij} \sim \text{Bin}(N_{ij}, \pi_{ij}) \quad i, j = 1, \dots, I \quad i \neq j$$

where each independent N_{ij} is the total number of comparisons between i and j . The PMF for just the pair (i, j) is hence

$$\begin{aligned} f_{ij}(n_{ij} \mid N_{ij}, \pi_{ij}) &= \binom{N_{ij}}{n_{ij}} \pi_{ij}^{n_{ij}} (1 - \pi_{ij})^{N_{ij} - n_{ij}} \\ \Rightarrow f_{ij}(n_{ij} \mid \beta_i, \beta_j, N_{ij}) &= \binom{N_{ij}}{n_{ij}} \left(\frac{\exp \beta_i}{\exp \beta_i + \exp \beta_j} \right)^{n_{ij}} \left(\frac{\exp \beta_j}{\exp \beta_i + \exp \beta_j} \right)^{N_{ij} - n_{ij}} \\ &= \binom{N_{ij}}{n_{ij}} \frac{(\exp \beta_i)^{n_{ij}} (\exp \beta_j)^{N_{ij} - n_{ij}}}{(\exp \beta_i + \exp \beta_j)^{N_{ij}}} \end{aligned}$$

Denote $\boldsymbol{\beta} = (\beta_1, \dots, \beta_I)^\top$ as the vector of all team strength parameters to estimate and $\mathcal{T}$ as the set of all paired observations. Plugging this expression into the likelihood for each unique pair (i, j) in the competition, we have

$$L(\boldsymbol{\beta} \mid \mathcal{T}) = \prod_{i < j} f_{ij}(n_{ij} \mid \beta_i, \beta_j, N_{ij}) = \prod_{i < j} \binom{N_{ij}}{n_{ij}} \frac{(\exp \beta_i)^{n_{ij}} (\exp \beta_j)^{N_{ij} - n_{ij}}}{(\exp \beta_i + \exp \beta_j)^{N_{ij}}}$$

where $\prod_{i < j}$ denotes the product over unique pairs (i, j) . Hence, the log-likelihood becomes

$$\begin{aligned} \log L(\boldsymbol{\beta} \mid \mathcal{T}) &= \sum_{i < j} \left[\log \binom{N_{ij}}{n_{ij}} + \log \frac{(\exp \beta_i)^{n_{ij}} (\exp \beta_j)^{N_{ij} - n_{ij}}}{(\exp \beta_i + \exp \beta_j)^{N_{ij}}} \right] \\ &\approx \sum_{i < j} [n_{ij} \beta_i + n_{ji} \beta_j - N_{ij} \log(\exp \beta_i + \exp \beta_j)] \end{aligned}$$

where we dropped the binomial coefficient term since we are concerned primarily with the kernel. We can go slightly further by recognising

$$\sum_{i < j} n_{ij} \beta_i = \sum_i n_{i+} \beta_i$$

where n_{i+} is the so-called ‘victory total’ for each $i = 1, \dots, I$.

The final expression for the log-likelihood is thus

$$\log L(\boldsymbol{\beta} \mid \mathcal{T}) = \sum_i n_{i+} \beta_i - \sum_{i < j} N_{ij} \log(\exp \beta_i + \exp \beta_j)$$

which shows that, given the $\{N_{ij}\}$, the victory totals $\{n_{ij}\}$ form a minimal sufficient statistic. As such, we can use MLE (via, for example, an EM algorithm) to find the parameter estimates $\hat{\beta}_i$ and hence the ranking of the I competitors.

Note: some sources get a slightly different expression for the log-likelihood, such as Hunter (2004), but the main result should be the same.